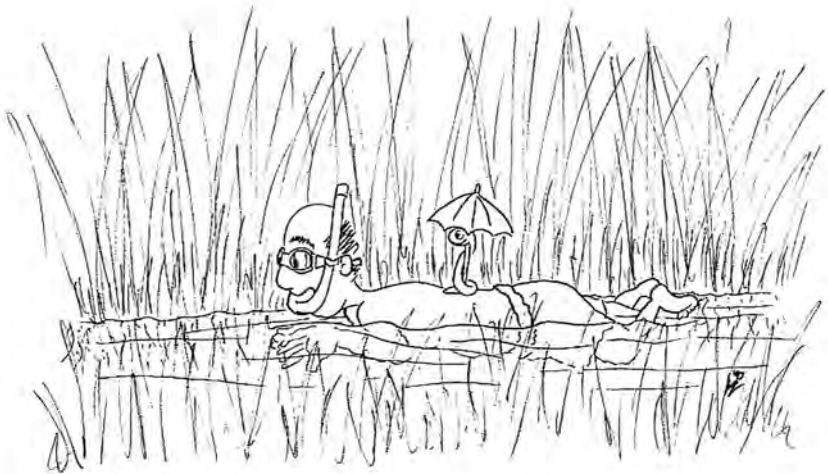


CHAPTER 3

Saturated Water Flow

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Introduction

Most hydrologic problems center on defining the amount of stored water and the rate and pathways of its movement through the subsurface. In particular, water resource problems focus on determining the amount of water that can be extracted from aquifers. Contaminant hydrogeology problems center on the movement of water and solutes toward wells or sensitive ecosystems. Ecohydrologic studies center on the role of water as part of natural systems, both as a primary resource and as a medium of nutrient transport. Porosity and compressibility define the amount of water that can be stored, while the energy gradient and permeability control the water flux. The major challenges of hydrologic science are to interpret the interactions of these characteristics, often with limited data. One major challenge facing subsurface hydrology is the technical difficulty of measuring hydrologic states and properties in situ in the subsurface; monitoring groundwater flow can also be expensive. A further complication in saturated flow research is that most measurements are based on small samples, while problems of interest often occur at larger scales. Given the inherent spatial variability of geologic materials, it can be difficult to know how to use field measurements for quantitative analyses.

In this chapter, we present problems of increasing difficulty through which the student is required to quantify the amount of stored soil water and determine the drainable fraction that can be released by compression, understand the relationships among pressure, pressure head, and hydraulic head, understand the relationships among permeability, gradient, flow, flux, and velocity and demonstrate knowledge of field applications of saturated flow problems.

References

Warrick, A.W. 2003. Soil Water Dynamics. Oxford University Press. 391p.

Symbols and Notation

A	Area	L^2
d	Height, depth	L
g	Gravity constant	$L T^{-2}$
h	Soil-water pressure head	L
h_z	Gravitational head	L
H	Total hydraulic head	L
k	Permeability	L^2
K	Hydraulic conductivity	$L T^{-1}$
K_s	Saturated hydraulic conductivity	$L T^{-1}$
K_{eq}	Equivalent saturated hydraulic conductivity	$L T^{-1}$

p	Pressure	$ML^{-1}T^{-2}$
q	Flux density	LT^{-1}
Q	Volumetric flow rate	L^3T^{-1}
Q_{2D}	Volumetric flow rate per unit of thickness	L^2T^{-1}
T	Time	T
v	Pore water velocity	LT^{-1}
V	Volume	L^3
θ	Volumetric water content	L^3L^{-3}
μ	Fluid dynamic viscosity	$ML^{-1}T^{-1}$
ρ	Fluid density	ML^{-3}

Problem 3-1

Choose the correct response for each of the following. Answers are given at the end.

a. The fundamental law for fluid flow through saturated porous media is

- 1) Archie's law
- 2) Newton's law
- 3) Poiseuille's law
- 4) Darcy's law

b. Permeability has dimensions of

- 1) Length per time
- 2) Length per time squared
- 3) Length squared
- 4) Length cubed

c. Flux (more precisely the flux density) is the

- 1) Volume of water flowing through a unit cross section
- 2) Product of the water flux and the solute density
- 3) Mass of water flowing through a cylindrical tube
- 4) Volume of water flowing through a unit cross section per unit time

d. The constant head method for measuring the saturated hydraulic conductivity is based on the following flow conditions:

- 1) Non steady, three dimensional unsaturated flow
- 2) Steady, one dimensional saturated flow
- 3) Steady, one dimensional unsaturated flow
- 4) Non steady, two dimensional saturated flow

e. In an isotropic medium, water flows

- 1) From high hydraulic head to low hydraulic head
- 2) From low hydraulic head to high hydraulic head
- 3) From high pressure to low pressure
- 4) Downhill

Answer:

4, 3, 4, 2 and 1

Problem 3-2

Consider a medium with a porosity of $0.45 \text{ cm}^3 \text{ cm}^{-3}$ filling a 1 m^3 container.

a. How much water is released if the water is drained entirely?

Answer:

$$V_{\text{total}} = 1 \text{ m}^3$$

$$V_{\text{pores}} = (1)(0.45) = 0.45 \text{ m}^3$$

b. How much water is released if some “residual” water cannot be removed, leaving a water content of $0.05 \text{ cm}^3 \text{ cm}^{-3}$?

Answer:

$$V_{\text{drainable}} = (1)(0.45 - 0.05) = 0.40 \text{ m}^3$$

c. How much water is released if the medium remains fully saturated, but the column is compressed, reducing the total volume by 1%? Assume incompressible grains, such that all of the compression is accomplished through a reduction of porosity. Also, assume that the residual water content is unchanged.

Answer:

$$V_{\text{pores}} = 0.45 - (1)(0.01) = 0.44 \text{ m}^3$$

$$V_{\text{drainable}} = (1)(0.44 - 0.05) = 0.39 \text{ m}^3$$

Problem 3-3

Consider a 100 cm long column with a cross sectional area of 10 cm^2 packed with a medium with a permeability of $(10)^{-12} \text{ m}^2$ with pressure heads at the top and bottom of 0 and 200 cm, respectively

a. What is the Darcy flux?

Answer:

We take the (arbitrary) datum elevation to be located at the base of the column and choose z to be vertically upward. Taking $z = z_2$ at the top and $z = z_1$ at the bottom gives

$$q = -K \frac{\Delta H}{\Delta z} = -K \frac{(h_{z_2} + h_2) - (h_{z_1} + h_1)}{z_2 - z_1} = -K \frac{(100 + 0) - (0 + 200)}{100 - 0} = K$$

$$K = \frac{\rho g k}{\mu} = \frac{(1000)(9.81)(10)^{-12}}{0.001} = 9.81(10)^{-6} \text{ m s}^{-1}$$

$$q = K = 9.81(10)^{-6} \text{ m s}^{-1}$$

The positive sign indicates flow in the positive direction of z , i.e., upwards.

b. What is the volumetric flow rate of water?

Answer:

$$Q = |q|A = 9.81(10)^{-6}(10)^{-3} = 9.81(10)^{-9} \text{ m}^3 \text{ s}^{-1}$$

c. If the medium has a porosity of $\phi = 0.33 \text{ cm}^3 \text{ cm}^{-3}$, then on average, how long does it take a particle of water to flow through the column?

Answer:

$$t = \frac{L}{(|q|/\phi)} = \frac{1}{9.81(10)^{-6}/0.33} = 3.36(10)^4 \text{ s}$$

Problem 3-4

Consider a 100 cm long column with a cross sectional area of 10 cm^2 packed with a medium with a permeability of $(10)^{-12} \text{ m}^2$ with pressure heads at the top and bottom of 200 and 0 cm, respectively.

a. What is the Darcy flux?

Answer:

We take the (arbitrary) datum elevation to be located at the base of the column and choose z to be vertically upward. Taking $z = z_2$ at the top and $z = z_1$ at the bottom gives

$$q = -K \frac{\Delta H}{\Delta z} = -K \frac{(h_{z_2} + h_2) - (h_{z_1} + h_1)}{z_2 - z_1} = -K \frac{(100 + 200) - (0 + 0)}{100 - 0} = -3K$$

$$K = \frac{\rho g k}{\mu} = \frac{(1000)(9.81)(10)^{-12}}{0.001} = 9.81(10)^{-6} \text{ m s}^{-1}$$

$$q = -3K = (3)9.81(10)^{-6} = -2.94(10)^{-5} \text{ m s}^{-1}$$

The negative sign means that the flow is in the negative direction of z , i.e., downwards.

b. What is the volumetric flow rate of water out of the base of the column?

Answer:

$$Q = |q|A = 2.94(10)^{-8} \text{ m}^3 \text{ s}^{-1}$$

c. If the medium has a porosity of $\phi = 0.33 \text{ cm}^3 \text{ cm}^{-3}$, then on average how long does it take a particle of water to flow through the column?

Answer:

$$t = \frac{L}{(|q|\phi)} = \frac{1}{2.94(10)^{-5}/0.33} = 1.13(10)^4 \text{ s}$$

Problem 3-5

Consider a 100 cm long column with a cross sectional area of 10 cm^2 packed with a medium with a permeability of $(10)^{-12} \text{ m}^2$ with pressure heads at the top and bottom of 0 and 200 cm, respectively. A compound is added to the water that decreases its density by 10% and increases its viscosity by 10%.

a. What is the Darcy flux?

We take the (arbitrary) datum elevation to be located at the base of the column and choose z to be vertically upward. Taking $z = z_2$ at the top and $z = z_1$ at the bottom gives

$$q = -K \frac{\Delta H}{\Delta z} = -K \frac{(h_{z_2} + h_2) - (h_{z_1} + h_1)}{z_2 - z_1} = -K \frac{(100 + 0) - (0 + 200)}{100 - 0} = K$$

$$K = \frac{\rho g k}{\mu} = \frac{(1000)(0.9)(9.81)(10)^{-12}}{(0.001)(1.1)} = 8.03(10)^{-6} \text{ m s}^{-1}$$

$$q = K = 8.03(10)^{-6} \text{ m s}^{-1}$$

The positive sign means that the flow is in the positive direction of z , i.e., upwards.

b. What is the volumetric flow rate of water?

Answer:

$$Q = |q|A = 8.03(10)^{-6} (10)^{-3} = 8.03(10)^{-9} \text{ m}^3 \text{ s}^{-1}$$

c. If the medium has a porosity of $\phi = 0.33 \text{ cm}^3 \text{ cm}^{-3}$, then on average, how long does it take a particle of water to flow through the column?

Answer:

$$t = \frac{L}{(|q| \phi)} = \frac{1}{8.03(10)^{-6} / 0.33} = 4.11(10)^4 \text{ s}$$

d. Which of the changes (density or viscosity) made the largest difference in the travel time of the water?

Answer:

If you vary density and viscosity independently by 10%, you will find that the density change (decreases K to 0.9 of the original value) has a slightly larger impact than the viscosity change (decreases K to $1/1.1 = 0.909$ of the original value).

Problem 3-6

Consider a 100 cm long column that is packed with a uniform medium with a hydraulic conductivity of $4(10)^{-5} \text{ cm s}^{-1}$. The pressure head is 0 cm at the base of the column. The steady downward flux through the column is $6(10)^{-5} \text{ cm s}^{-1}$. The volumetric water content is equal to the porosity and is $0.45 \text{ cm}^3 \text{ cm}^{-3}$.

a. What is the hydraulic head at the top of the column?

Answer:

We take the (arbitrary) datum elevation to be located at the base of the column and choose z to be vertically upward given $q = -6(10)^{-5} \text{ cm s}^{-1}$.

Taking $z = z_2$ at the top and $z = z_1$ at the bottom gives

$$q = -K \frac{\Delta H}{\Delta z} = -K \frac{H_2 - H_1}{z_2 - z_1}$$

$$H_2 = \frac{q(z_2 - z_1)}{-K} + H_1 = \frac{-6(10)^{-5}(100 - 0)}{-4(10)^{-5}} + 0 = 150 \text{ cm}$$

b. Find the pressure head, h , at 20 cm above the base of the column.

Answer:

$$H_{20} = \frac{q(z_2 - z_1)}{-K} + H_1 = \frac{-6(10)^{-5}(20 - 0)}{-4(10)^{-5}} + 0 = 30 \text{ cm}$$

$$h_{20} = H_{20} - h_{z,20} = 30 - 20 = 10 \text{ cm}$$

c. Calculate the average residence time of water in the column during steady state flow.

Answer:

$$t = \frac{L}{(|q|/\phi)} = \frac{1}{6(10)^{-5}/0.45} = 7.5(10)^3 \text{ s}$$

Problem 3-7

The gauge pressure of water in a soil is reported as 0.2 atmospheres. The pressure was measured 3 m above the local datum elevation. A second pressure measurement at an elevation of 10 m above the local datum and 4.0 m laterally from the first point also showed a pressure head of 0.2 atmospheres. The medium is homogeneous and isotropic with an intrinsic permeability of $(10)^{-10} \text{ m}^2$.

a. What is the pressure head, h , at both locations in meters of water?

Answer:

One atmosphere is equal to $1.01(10)^5 \text{ Pa}$ which is the mks unit of pressure.

$$p = 0.2(1.01)(10)^5 = 2.02(10)^4 \text{ Pa}$$

$$h = \frac{p}{\rho g} = \frac{(2.02)(10)^4}{(1000)(9.81)} = 2.06 \text{ m}$$

b. Find the hydraulic head, H at each location in meters of water.

Answer:

$$H_1 = h_1 + h_{z1} = 2.06 + 3 = 5.06 \text{ m}$$

$$H_2 = h_2 + h_{z2} = 2.06 + 10 = 12.06 \text{ m}$$

c. What is the hydraulic conductivity of the medium expressed in units of m s^{-1} ? Assume the viscosity is $\mu = 0.001 \text{ kg m}^{-1} \text{ s}^{-1}$.

Answer:

$$K = \frac{\rho g k}{\mu} = \frac{(1000)(9.81)(10)^{-10}}{0.001} = 9.81(10)^{-4} \text{ m s}^{-1}$$

d. Use Darcy's law to find the magnitude of the water flux, $|q|$.

Answer:

First calculate the distance L between the two points

$$L = \left[4^2 + (10 - 3)^2 \right]^{0.5} = 8.06 \text{ m}$$

Then the flow becomes

$$|q| = K \frac{\Delta H}{L} = K \frac{H_2 - H_1}{L} = 9.81(10)^{-4} \frac{12.06 - 5.05}{8.06} = 8.52(10)^{-4} \text{ m s}^{-1}$$

The flow occurs from the high to low hydraulic head, i.e., from the second to the first point.

e. Determine whether flow is laminar or turbulent given that the porosity (ϕ) is $0.40 \text{ cm}^3 \text{ cm}^{-3}$ and that the d_{50} median grain diameter is 0.10 cm .

Answer:

Flow is considered laminar if Reynolds' number, R (dimensionless) is less than 10.

$$R = \frac{\rho |q| d_{50}}{\mu \phi} = \frac{(1000)(8.52)(10)^{-4} (0.001)}{(0.001)(0.4)} = 2.13$$

Flow is laminar for these conditions.

Problem 3-8

Consider two piezometers installed in a saline aquifer. One piezometer is screened at the local datum elevation. The other is screened 10 m above the datum and 100 m horizontally from the first. The water level in the

first aquifer is 10 m above the base of the well. The water level in the second is 25 m above the base of the well. The water has a density of 1100 kg m^{-3} . The medium is homogeneous and isotropic with an intrinsic permeability of $(10)^{-10} \text{ m}^2$.

a. What is the pressure difference between the two monitored locations?

Answer:

$$p_1 = \rho g h_1 = (1100)(9.81)(10) = 1.08(10)^5 \text{ Pa}$$

$$p_2 = \rho g h_2 = (1100)(9.81)(25) = 2.7(10)^5 \text{ Pa}$$

$$p_2 - p_1 = 2.7(10)^5 - 1.08(10)^5 = 1.62(10)^5 \text{ Pa}$$

b. What is the magnitude of the Darcy flux between the piezometers?

Answer:

$$K = \frac{\rho g k}{\mu} = \frac{(1100)(9.81)(10)^{-10}}{0.001} = 1.08(10)^{-3} \text{ m s}^{-1}$$

$$L = (100^2 + 10^2)^{0.5} = 100.5 \text{ m}$$

$$|q| = K \frac{\Delta H}{L} = K \frac{(h_2 + h_{z2}) - (h_1 + h_{z1})}{L} = 1.08(10)^{-3} \frac{(25 + 10) - (10 + 0)}{100.5} = 2.69(10)^{-4} \text{ m s}^{-1}$$

The flow occurs from the high to low hydraulic head, i.e., from the second to the first point.

c. Is flow laminar?

Answer:

Assume that the d_{50} median grain diameter is 0.10 cm and calculate the Reynolds' number, R (dimensionless).

$$R = \frac{\rho |q| d_{50}}{\mu \phi} = \frac{(1100)(2.7)(10)^{-4} (0.001)}{(0.001)(0.4)} = 0.74$$

Flow is laminar for these conditions ($R < 10$).

Problem 3-9

The following pressure head measurements were made in a 100 cm long vertical soil column. At the base of the column the pressure head was 0 cm; at the top of the column the pressure head was 10 cm; 25 cm above

the base of the column, the pressure head was 40 cm. The column is undergoing steady state flow.

Plot the gravitational head, pressure head, and hydraulic head (x-axis) as a function of elevation above the base of the column (y-axis) for the three measurement locations. Define the direction of flow.

Answer:

Choose the bottom of the column to be the datum elevation ($z = 0$). The elevation head increases 1:1 with elevation. The pressure head values are given. The hydraulic head, H at each elevation is equal to the sum of the pressure head, h and gravitational head h_z .

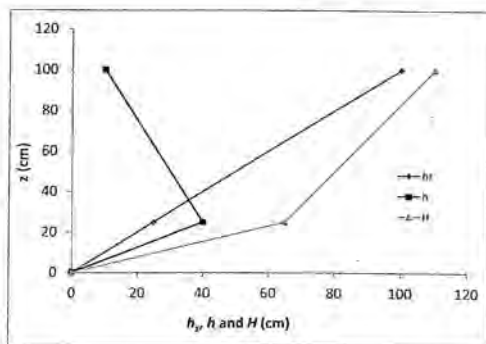


Fig. 3-1: Head components as a function of elevation

The hydraulic head is highest at the top of the column, so flow is downward.

b. Explain why there is a change in slope in the hydraulic head profile from the bottom to the middle location compare to the slope from the middle to the top location.

Answer:

For one-dimensional steady-state flow, this change in gradient must be associated with a change in the hydraulic conductivity, as the Darcy flux is constant. Specifically, the magnitude of the gradient (change in head per change in elevation) is higher in the lower part of the column, indicating that the material in this lower region has a lower hydraulic conductivity than the material in the upper region.

c. Determine the height of ponding.

Answer:

The pressure head at the top of the column is 10 cm. This indicates that 10 cm of water is ponded at the top of the column.

d. Explain why you cannot determine the flux through the column from the figure. Describe one other piece of information that would allow you to determine the flux.

Answer:

The flux is the product of the hydraulic head gradient and the hydraulic conductivity. The flux could be defined if the hydraulic conductivity in the two layers of the column and the elevation of the layer of contact between them was known. Alternatively, the flux could be determined from the top and bottom hydraulic head values if the effective hydraulic conductivity of the column was known.

Problem 3-10

A 100 cm long column is packed with a uniform medium with a hydraulic conductivity of $4(10)^{-5} \text{ cm s}^{-1}$. After packing, the uppermost 1 cm is replaced with a medium with a hydraulic conductivity of $2.2(10)^{-7} \text{ cm s}^{-1}$. The pressure head is 20 cm at the base of the column and 150 cm at the top of the column. The volumetric water content is equal to the porosity and is $0.45 \text{ cm}^3 \text{ cm}^{-3}$. Set the datum elevation at the base of the column.

a. Find the equivalent vertical hydraulic conductivity of the column.

Answer:

The equivalent or effective conductivity is (cf., Warrick, 2003, Eq. 5-144)

$$K_{eq} = \frac{L_{upper} + L_{lower}}{\frac{L_{upper}}{K_{upper}} + \frac{L_{lower}}{K_{lower}}} = \frac{1 + 99}{\frac{1}{2.2(10)^{-7}} + \frac{99}{4(10)^{-5}}} = 1.42(10)^{-5} \text{ cm s}^{-1}$$

b. Determine the steady state flow through the column.

Answer:

$$q = -K_{eq} \frac{\Delta H}{\Delta z} = -1.42(10)^{-5} \frac{(100 + 150) - (0 + 20)}{100 - 0} = -3.27(10)^{-5} \text{ cm s}^{-1}$$

(Flow is from higher to lower hydraulic head, i.e., from the top to the bottom).

c. What is the average residence time of water in the column during steady state flow?

Answer:

$$t = \frac{L}{(|q|/\phi)} = \frac{100}{3.27(10)^{-5}/0.45} = 1.38(10)^6 \text{ s}$$

d. Determine the pressure head, h_b at the interface between the two layers.

Answer:

In the lower layer Δz is 99 cm resulting in

$$\Delta H = \frac{-q\Delta z}{K_{\text{lower}}} = \frac{3.27(10)^{-5}(99)}{4(10)^{-5}} = 80.93 \text{ cm}$$

$$H_b = \Delta H + H_{\text{lower}} = 80.93 + 20 = 100.93 \text{ cm}$$

$$h_b = H_b - h_{zb} = 100.93 - 99 = 1.93 \text{ cm}$$

e. Explain why water flows from lower pressure at the boundary between the layers to higher pressure at the base of the column.

Answer:

The hydraulic head gradient controls flow, not the pressure head gradient. In this case, the difference in elevation is sufficient to overcome the greater pressure at the base of the column, leading to downward flow from a region of low pressure to a region of high pressure.

f. How high would water rise in a piezometer inserted 50 cm above the base of the column.

Answer:

$$\Delta H = \frac{-q\Delta z}{K_{\text{lower}}} = \frac{3.27(10)^{-5}(50)}{4(10)^{-5}} = 40.88 \text{ cm}$$

$$H_{50} = \Delta H + H_{\text{lower}} = 40.88 + 20 = 60.88 \text{ cm}$$

$$h_{50} = H_{50} - h_{z50} = 60.88 - 50 = 10.88 \text{ cm}$$

The water would rise to a height of 60.88 cm above the datum, which is equivalent to a height of 10.88 cm above the bottom of the piezometer.

Problem 3-11

A 100 cm long column is packed with two different soils. The pressure head at the base of the column is 0 cm. The pressure head at a boundary between two soil layers, located 25 cm above the base of the column, is 40 cm. The upper layer has a hydraulic conductivity of $2.55(10)^{-5} \text{ cm s}^{-1}$. The steady flux in the column is $q = -1.2(10)^{-5} \text{ cm s}^{-1}$.

a. What is the K of the lower layer?

Answer:

$$K_{\text{lower}} = -\frac{q}{\Delta H / \Delta z} = -\frac{-1.2(10)^{-5}}{\left[\frac{(40 + 25) - (0 + 0)}{(25 - 0)} \right]} = 4.62(10)^{-6} \text{ cm s}^{-1}$$

b. What is the gradient, $\Delta H / \Delta z$ in the upper layer?

Answer:

$$\frac{\Delta H}{\Delta z} = -\frac{q}{K_{\text{upper}}} = -\frac{-1.2(10)^{-5}}{2.55(10)^{-5}} = 0.47 \text{ cm cm}^{-1}$$

c. What is the height of ponded water at the top of the column?

Answer:

$$H_{100} = H_b + 0.47 \Delta z = (40 + 25) + (0.47)(75) = 100.25 \text{ cm}$$

$$h_{100} = H_{100} - h_{z100} = 100.25 - 100 = 0.25 \text{ cm}$$

The water is ponded to a depth of 0.25 cm at the top of the column.

d. What is the equivalent hydraulic conductivity of the medium?

Answer:

$$K_{\text{eq}} = -\frac{q}{\frac{\Delta H}{\Delta z}} = -\frac{-1.2(10)^{-5}}{\left[\frac{(0.25 + 100) - (0 + 0)}{(100 - 0)} \right]} = 1.2(10)^{-5} \text{ cm s}^{-1}$$

Problem 3-12

A horizontal column of 100 cm is packed with a uniform medium with a hydraulic conductivity of $4(10)^{-5} \text{ cm s}^{-1}$. The leftmost 1 cm is replaced with a medium with a hydraulic conductivity of $2.2(10)^{-7} \text{ cm s}^{-1}$. The pressure head is 20 cm at the left end of the column and 150 cm at the right end. The volumetric water content is equal to the porosity and is $0.45 \text{ cm}^3 \text{ cm}^{-3}$. Flow is horizontal and steady state.

a. Determine the equivalent horizontal hydraulic conductivity of the column.

Answer:

Using the same expression as in problem 3-10, K_{eq} is:

$$K_{eq} = \frac{L_{left} + L_{right}}{\frac{L_{left}}{K_{left}} + \frac{L_{right}}{K_{right}}} = \frac{1 + 99}{\frac{1}{2.2(10)^{-7}} + \frac{99}{4(10)^{-5}}} = 1.42(10)^{-5} \text{ cm s}^{-1}$$

b. Calculate the flow through the column taking $x = 0$ at the left and $x = 100 \text{ cm}$ at the right.

Answer:

$$q = -K_{eq} \frac{\Delta H}{\Delta x} = -1.42(10)^{-5} \frac{(150 + 0) - (20 + 0)}{100 - 0} = -1.85(10)^{-5} \text{ cm s}^{-1}$$

Flow occurs from right to left, i.e., in the negative direction. We have assumed the gravitational head to be zero throughout.

c. Find the average residence time of water in the column.

Answer:

$$t = \frac{L}{(|q|/\phi)} = \frac{100}{1.85(10)^{-5}/0.45} = 2.43(10)^6 \text{ s}$$

d. Find the pressure head at the interface between the two layers:

Answer:

$$\Delta H = \frac{-q\Delta x}{K_{left}} = \frac{-1.85(10)^{-5}(1)}{2.2(10)^{-7}} = 84 \text{ cm}$$

$$H_b = \Delta H + H_{left} = 84 + 20 = 104 \text{ cm}$$

$$h_b = H_b - h_{z_b} = 104 - 0 = 104 \text{ cm}$$

e. Explain why, unlike the same column with the same conditions placed vertically in a previous question, water in the horizontal column flows out of the lower pressure boundary.

Answer:

In the absence of an elevation head gradient (horizontal flow), the direction of the hydraulic head gradient is the same as the direction of the pressure head gradient. Therefore, in this case, flow occurs from right to left and is from the higher to lower pressure.

f. Calculate the elevation, relative to the mid-elevation of the column, to which water would rise in a piezometer inserted 50 cm from the right end of the column:

Answer:

Apply Darcy's law from the midway point to the right end of the column

$$\Delta H = \frac{-q\Delta x}{K_{right}} = -\frac{-1.85(10)^{-5}(50)}{4(10)^{-5}} = 23.25 \text{ cm}$$

$$H_{50} = H_{right} - \Delta H = (150 + 0) - 23.25 = 126.75 \text{ cm}$$

$$h_{50} = H_{50} - h_{z50} = 126.75 - 0 = 126.75 \text{ cm}$$

Problem 3-13

A horizontal column of 100 cm length and 10 cm height is packed with a uniform medium with a hydraulic conductivity of $4(10)^{-5} \text{ cm s}^{-1}$. Then, the uppermost 1 cm is replaced with a medium with a hydraulic conductivity of $2.2(10)^{-7} \text{ cm s}^{-1}$ (The horizontal column has two horizontal layers). The pressure head is 20 cm at the left end of the column and 150 cm at the right end. The volumetric water content of both soils is equal to the porosity and is 0.45 cm cm^{-1} . Flow is horizontal and steady state.

a. Determine the equivalent horizontal hydraulic conductivity of the column.

Answer:

Using the expression for K_{eq} for flow in parallel:

$$K_{eq} = \frac{L_{top}K_{top} + L_{bottom}K_{bottom}}{L_{top} + L_{bottom}} = \frac{(1)2.2(10)^{-7} + (9)4(10)^{-5}}{1 + 9} = 3.6(10)^{-5} \text{ cm s}^{-1}$$

b. Calculate the flow through the column taking $x = 0$ at the left and $x = 100$ cm at the right.

Answer:

$$q = -K_{eq} \frac{\Delta H}{\Delta x} = -3.6(10)^{-5} \frac{(150+0) - (20+0)}{100-0} = -4.68(10)^{-5} \text{ cm s}^{-1}$$

c. Find the flux in the top and bottom layers of the column.

Answer:

$$q_{top} = -K_{top} \frac{\Delta H}{\Delta x} = -2.2(10)^{-7} \frac{(150+0) - (20+0)}{100-0} = -2.86(10)^{-7} \text{ cm s}^{-1}$$

$$q_{bottom} = -K_{bottom} \frac{\Delta H}{\Delta x} = -4(10)^{-5} \frac{(150+0) - (20+0)}{100-0} = -5.2(10)^{-5} \text{ cm s}^{-1}$$

d. Find the average residence time of water in the top and bottom layers of the column.

Answer:

$$t_{top} = \frac{L}{(|q_{top}| / \phi)} = \frac{100}{2.86(10)^{-7} / 0.45} = 1.57(10)^8 \text{ s}$$

$$t_{bottom} = \frac{L}{(|q_{bottom}| / \phi)} = \frac{100}{5.2(10)^{-5} / 0.45} = 8.65(10)^5 \text{ s}$$

Problem 3-14

The figure below shows steady state flow lines in an unconfined aquifer. The dashed line shows the water table elevation. Four piezometers are shown, each of which is only open at its base (no screen).

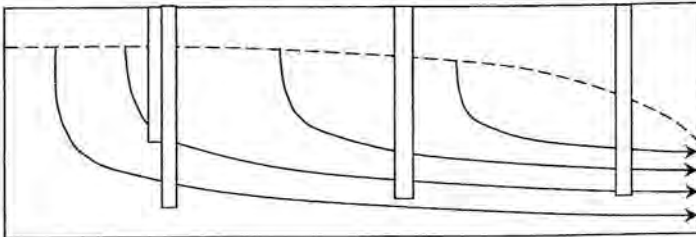


Fig. 3-2: Steady state flow lines in an unconfined aquifer

a. For each piezometer, sketch an equipotential (line of equal hydraulic head) which intersects both the lower opening of the piezometer and the water table.

Answer:

Equipotentials can be drawn through any point and continued orthogonal to the flow lines. Note that equipotentials are not necessarily orthogonal to the water table (See Figure 3-3).

b. Based on the definition of the water table, graphically determine the water levels and show them on the sketch.

Answer:

The water table is defined as the collection of points at which the water pressure is equal to zero. Therefore, the hydraulic head within each piezometer, is equal to the water column height above an arbitrary common datum elevation. The water will rise to the height of the water table at the point of intersection with the equipotential that passes through the intake of the piezometer.

The results can be shown as the water level in each piezometer compared to the water table elevation. Note that the water level in the well is consistently lower than the water table, indicating downward gradients across the water table, which are consistent with the direction of flow recharging the unconfined aquifer.

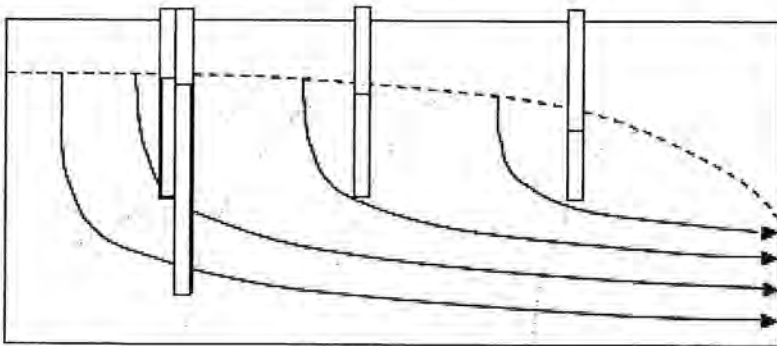


Fig. 3-3: Equipotentials and piezometers in an unconfined aquifer

c. How would the water levels in the well change if they were screened over their entire length?

Answer:

It is possible that the vertical gradients in the aquifer would cause downward flow in the wells. But, ignoring this possibility, as we commonly

do, the water in the piezometers would rise to the elevation of the water table where the well intersects the water table.

Problem 3-15

A wastewater treatment plant uses water filtration as a final treatment phase. They have an infiltration basin with an area of $A = 600 \text{ m}^2$. Water is removed at a depth of 1.0 m using closely-spaced tile drains, which maintain a zero pressure head at that depth. The saturated hydraulic conductivity of the medium is 1.2 m d^{-1} .

a. Calculate the ponding height needed to treat 950 m^3 of wastewater per day assuming that the wastewater has a density of 1.0 g cm^{-3} .

Answer:

Set the datum at the tile drains and assume z positive upward.

$$|q| = \frac{Q}{A} = \frac{950}{600} = 1.583 \text{ m d}^{-1}$$

$$\Delta H = -\frac{q\Delta z}{K} = -\frac{(-1.583)(1-0)}{1.2} = 1.32 \text{ m}$$

$$H_{\text{ground}} = \Delta H + H_{\text{tile}} = 1.32 + (0+0) = 1.32 \text{ m}$$

$$h_{\text{ground}} = H_{\text{ground}} - h_{\text{zground}} = 1.32 - 1 = 0.32 \text{ m}$$

It would require 0.32 m of ponding to achieve this flow rate.

b. With time, fine particles and biological material accumulate at the ground surface. Calculate the ponding height that would be needed if the hydraulic conductivity of the upper 1 cm of the medium was reduced by two orders of magnitude.

Answer:

The equivalent hydraulic conductivity of the medium is written below as in previous problems:

$$K_{eq} = \frac{\frac{L_{\text{top}}}{K_{\text{top}}} + \frac{L_{\text{bottom}}}{K_{\text{bottom}}}}{\frac{L_{\text{top}}}{K_{\text{top}}} + \frac{L_{\text{bottom}}}{K_{\text{bottom}}}} = \frac{\frac{0.01}{0.012} + \frac{0.99}{1.2}}{\frac{0.01}{0.012} + \frac{0.99}{1.2}} = 0.6 \text{ m d}^{-1}$$

$$\Delta H = -\frac{q\Delta z}{K} = -\frac{(-1.583)(1-0)}{0.6} = 2.64 \text{ m}$$

$$H_{\text{ground}} = \Delta H + H_{\text{tile}} = 2.64 + (0 + 0) = 2.64 \text{ m}$$

$$h_{\text{ground}} = H_{\text{ground}} - h_{\text{zground}} = 2.64 - 1 = 1.64 \text{ m}$$

It would require 1.64 m of ponding to achieve this flow rate, which is logical, as the hydraulic conductivity is now much lower at the surface.

c. Calculate how much the hydraulic conductivity would change compared to that for clean water if the viscosity and density of the wastewater were each 10% higher than for clean water.

Answer:

$$K = \frac{\rho g k}{\mu}$$

The changes offset one another exactly; there is no change in the hydraulic conductivity which remains at 1.2 m d^{-1} .

Problem 3-16

Consider a flownet that has six stream tubes crossing the right boundary of the domain. The gradient causes water to flow out of the domain. The head contour interval is 1 m and the cell widths and lengths are approximately equal. The saturated hydraulic conductivity is 1.2 m d^{-1} , and the porosity is $0.4 \text{ m}^3 \text{ m}^{-3}$.

a. What is the total flow out of the right boundary of the domain?

Answer:

The volumetric flow rate for a given stream tube per unit thickness is ΔQ_{2D} given by

$$\Delta Q_{2D} = K \left| \frac{\Delta H}{\Delta s} \right| \Delta w = (1.2)(1) = 1.2 \text{ m}^2 \text{ d}^{-1}$$

where the length of a flow cell in the direction of flow Δs is equal to the width of the cell Δw . As the flow is equal in each of the 6 stream tubes, the total flow out the right boundary is

$$Q_{2D} = 6\Delta Q_{2D} = 6(1.2) = 7.2 \text{ m}^2 \text{ d}^{-1}$$

b. Two points lie along the centerline of a flow tube and are separated by a distance of $L = 100 \text{ m}$. The flow tube width, Δw is 5 m. What is the travel time of a water molecule between the points?

Answer:

$$t = \frac{L\phi}{(\Delta Q_{2D}/\Delta w)} = \frac{100(0.4)}{(1.2/5)} = 167 \text{ d}$$

c. How can you determine if a water molecule located somewhere in the domain will exit the right boundary?

Answer:

Water molecules that are located within flow tubes that flow out across the boundary will exit the right boundary. All others will not.

Problem 3-17

Consider a flownet that includes a pumping well fed from a perennial stream. The saturated hydraulic conductivity is 1.2 m d^{-1} , the aquifer thickness, b is 20 m, and the porosity is $0.4 \text{ m}^3 \text{ m}^{-3}$:

- Ten flow tubes terminate at the well (Figure 3-4). The head contour between the equipotential lines is 10 m. What is the total flow into the well?
-

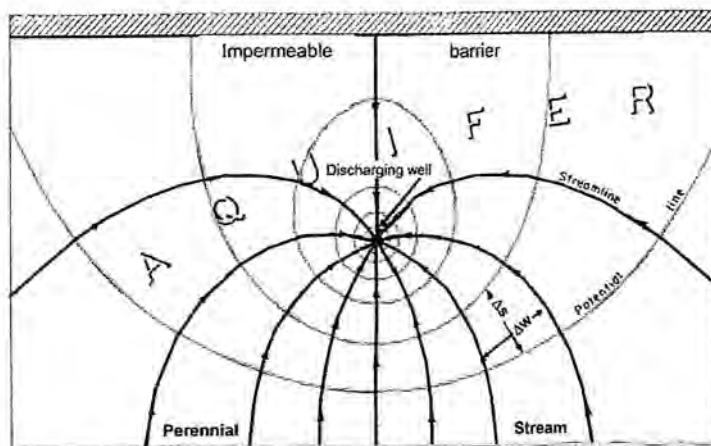


Fig. 3-4: Flownet for pumping well (plan view)

Answer:

The flow per unit thickness in each stream tube is

$$\Delta Q_{2D} = K \left| \frac{\Delta H}{\Delta s} \right| \Delta w = (1.2)(10) = 12 \text{ m}^2 \text{ d}^{-1}$$

with the length of a flow cell in the direction of flow Δs equal to the width of the cell Δw (See Figure 3-4). Thus, multiplication by 10 for the total number of stream tubes and by the thickness $b = 20 \text{ m}$ gives the total flow into the well:

$$Q = n\Delta Q_{2D}b = 10(12)(20) = 2400 \text{ m}^3 \text{ d}^{-1}$$

b. How can you identify the capture zone of the well?

Answer:

All flow lines that terminate in the well are part of the capture zone. In this case, the entire domain is captured by the well.

c. How would you calculate the travel time from any point on the boundary to the well?

Answer:

For each cell in the domain, the travel time can be calculated by

$$t = \frac{(\Delta s)\phi}{(\Delta Q_{2D}/\Delta w)} = \frac{(\Delta s)^2\phi}{\Delta Q_{2D}}$$

where ΔQ_{2D} the volumetric flow rate per unit thickness is the same for all of the cells and the cell length Δs remains equal to the cell width Δw . Beginning on the boundary, calculate the travel time across the first cell. Repeat this for the next cell in the flow tube until you reach the well. Then add the travel times to get the total time to the well. For example, consider one of the flow tubes at the center of the bottom boundary. If the width and height of the first cell are 100 m, the travel time would be

$$t = \frac{(\Delta s)^2\phi}{\Delta Q_{2D}} = \frac{(100)^2(0.4)}{12} = 333 \text{ d}$$

The next cell may be 80 m by 80 m. The travel time through this cell is

$$t = \frac{(\Delta s)^2\phi}{\Delta Q_{2D}} = \frac{(80)^2(0.4)}{12} = 213 \text{ d}$$

If the next cells are 60, 40, 20, and 10 m on a side, their travel times are 120, 53.3, 13.3 and 3.33 d resulting in a total travel time of 737 d.

Problem 3-18

Consider a 100 cm long vertical column. The hydraulic head at the top of the column is initially 150 cm. This is achieved by establishing a 50 cm ponded height above the column in a reservoir with the same cross sectional area as the column. The hydraulic head at the bottom is maintained at 0 cm. The saturated hydraulic conductivity is 1 cm d^{-1} .

a. What is the initial flux from the base of the column?

Answer:

$$q = -K \frac{\Delta H}{\Delta z} = -1 \frac{150 - 0}{100 - 0} = -1.5 \text{ cm d}^{-1}$$

The negative sign means that the flow is in the negative direction of z , i.e., downwards.

b. Assume that no more water is added to the top of the column, but water flows out of the base of the column at this rate for one day. What would the ponded height be at the end of the first day, h_{top}^* ?

Answer:

The flux describes the rate at which the water level in the reservoir is dropping. Therefore, after one day, the ponded height would be:

$$H_{\text{top}}^* = H_{\text{top}} - |q|t = 150 - 1.5(1) = 148.5 \text{ cm}$$

$$h_{\text{top}}^* = H_{\text{top}}^* - h_{\text{ztop}}^* = 148.5 - 100 = 48.5 \text{ cm}$$

c. Given the new ponded height, what would the flux be for the second day?

Answer:

$$q = -K \frac{\Delta H}{\Delta z} = -1 \frac{148.5 - 0}{100 - 0} = -1.48 \text{ cm d}^{-1}$$

d. Repeat this day-by-day analysis until the ponded height is zero and plot the ponded height versus time.

Answer:

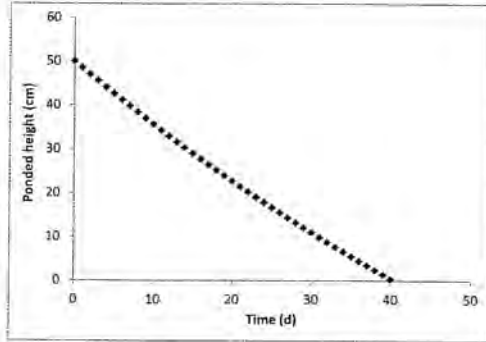


Fig. 3-5: Pondered height versus time

e. Discuss why the points that you plotted would not exactly predict the flux out of the column with time.

Answer:

The pondered height (and therefore, the flux) is actually changing continuously, and not in daily increments. Therefore, the flux is over-estimated, which means that the time required for the ponding height to decrease to zero is underestimated.

(The exact answer is 40.55 d as may be found using the results of Problem 5-15).

Problem 3-19

Consider the 100 cm long vertical column analyzed in the previous question.

a. If the porosity of the medium is $0.33 \text{ cm}^3 \text{ cm}^{-3}$, how would the average pore water velocity vary with time?

Answer:

Ignoring direction the water velocity, $v = \frac{|q|}{\phi}$ would be a constant multiple of the flux and would show the same type of nonlinear change with time:

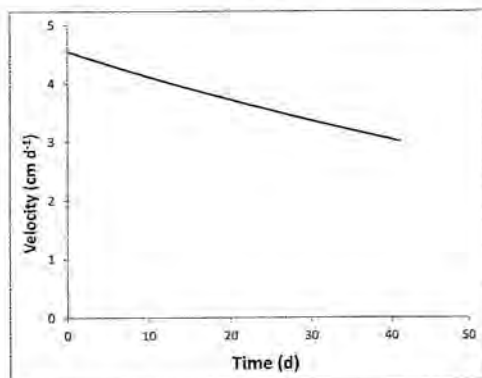


Fig. 3-6: Pore water velocity with time

b. Consider a water particle that begins at the top of the soil column. Plot its elevation with time until it reaches the lower end of the column.

Answer:

For each day, the water molecule would be displaced by the velocity multiplied by one day until it would reach the bottom of the column (approximately after 25 days).

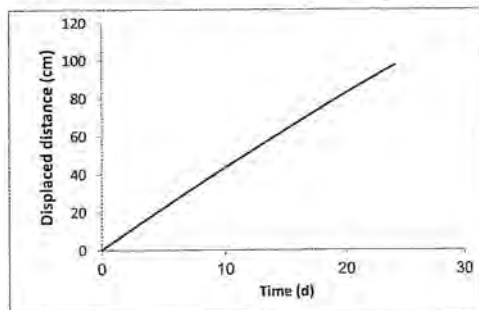


Fig. 3-7: Displaced distance versus time

c. What length of water has been infiltrated into the column when the particle considered above has reached the bottom of the column?

Answer:

From the figure above, the water molecule will have moved 100 cm in approximately 25 days. At this time, the ponded height can be estimated at 16.67 cm, which means that $50 - 16.67 = 33.3$ cm of water has been added to the column.

d. Explain why the length calculated above does not equal the length of the column.

Answer:

As water moves through the column it only displaces water in soil pores.

In theory, ϕL water should be added in order to displace one pore volume. This would be 33 cm for the porosity given, which is close to the result from our discrete daily calculations.

Problem 3-20

Consider a 100 cm long vertical column. The hydraulic head at the top of the column is initially 150 cm. This is achieved by establishing a 50 cm ponded height above the column in a reservoir with the same cross sectional area as the column. The hydraulic head at the bottom is maintained at 0 cm. The saturated hydraulic conductivity is 1 cm d^{-1} .

a. In the previous problem, you calculated the 'step-wise' flux from the base of the column. Now calculate the average pressure head within the column and plot it as a function of time.

Answer:

We are treating each step as steady state. For the homogeneous column, the pressure head varies linearly with elevation. Therefore, for each time step, the average pressure head is half of the pressure head at the top of the column (which equals the ponded height), as the pressure head is fixed to zero at the base of the column. For detailed calculations see EXCEL file.

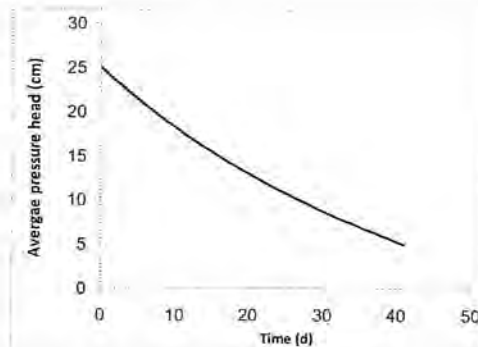


Fig. 3-8: Average pressure head versus time

b. Now assume that the medium is highly compressible, such that the hydraulic conductivity decreases with decreasing pressure head. Assume a linear dependence: $K_s = 0.02h + 0.5$, where h is the average pressure head in the column. For this condition, recalculate the ponded height as a function of time. Plot this ponded height and the ponded height for an incompressible medium versus time on the same axes.

Answer:

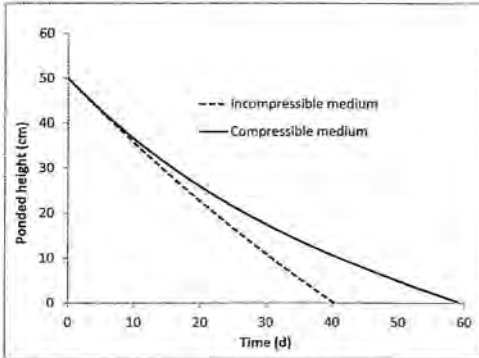


Fig. 3-9: Pondered height versus time

c. For the compressible medium, assume that the porosity changes as

$$\phi = \frac{0.33h}{125} + 0.264$$

Recalculate the velocity as a function of time. Plot the travel distance for both media together, as a function of time. Truncate each series at 100 cm. Discuss the differences among the series.

Answer:

If the porosity decreases as the flux decreases, then the effect of compression on the travel distance is mediated, when compared alongside the travel distance as calculated assuming a constant porosity but decreasing K . The former state is also the more realistic state, as compression would naturally decrease the porosity of the medium.

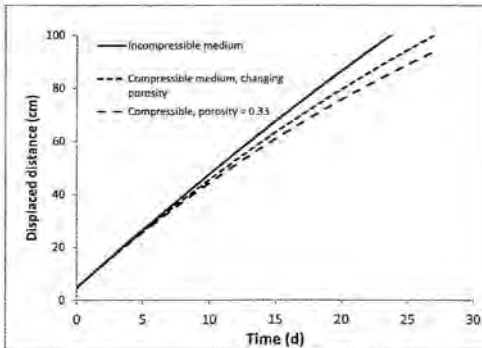


Fig. 3-10: Displaced distance as a function of time



If we knew what it was we were doing, it would not be called research, would it - Albert Einstein